

COLLEGE OF ENGINEERING, PUNE
(An Autonomous Institute of Govt. of Maharashtra)

End Semester Examination – November, 2013

(CT 205) Data Structures

Class: - T.Y. B.Tech (Computer Engineering & Information Technology)

Year: - 2013-14

Semester: - III

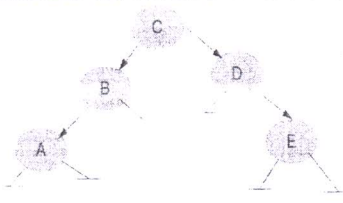
Duration: - Max 5 Hours

Max. Marks: - 60

Instructions:

1. All questions are compulsory.
2. This is an open book test. You can read your textbooks/notes, but exchange is not allowed.
3. Some questions may not have definite answers.
4. You can make assumptions if required, but you should state them.
5. Code must be indented, commented and written in ANSI C and looking clean. You can write the code with a bold pencil.
6. You can explain your answer in Marathi/Hindi.

A	Draw diagram of the data structure created by following code: <pre>int main() { char a[6], *p, *q; p = a + 3; q = a + (p - a) + 2; q[-2] = 'c'; p[-1] = 'd'; }</pre>	2
B	What will be the output of the following program? <pre>int main() { int i = 11, j = 10; printf("%d\n", i j); /* is bitwise OR */ }</pre>	2
C	Derive the equation for time taken by following function: <pre>int f(int n) { int i, j, x; for(i = 0; i < n/2; i+= 2) for(j = 0; j < n; j+= 4) x = i + j; }</pre>	2
D	Write a program, which opens a file (filename given as command line argument) and prints the contents of the file on screen.	2
E	Write a function which returns the index of first occurrence of a given character, in a string. <pre>int chridx(char *string, char ch);</pre> <i>For example: if string="somethinges" and ch='e' then the function returns 3.</i>	2
F	Write max selection sort for an array of characters. <pre>void selectsort(char a[], int n);</pre> <i>// 'n' is the size of array 'a'</i>	2
G	Write the prefix expression for the following infix expression $(((1 + 2) * (3 + 4)) / (5 / 6))$	2

	H	Draw a diagram of a singly-linked NULL terminated linked list, after following operations on it: <pre> init(&l); insert(&l, 0, 2); // insert at position 0, value 2 append(&l, 5); insert(&l, 1, 4); remove(&l, 0); // remove from position 0 append(&l, 10); insert(&l, 2, 6); </pre>	2
	I	Show the state of the following sequence of numbers, stored in an array, after each iteration of improved bubble sort. (assume alphabetical ordering for comparison) 100 50 30 20 40 60 80	2
	J	Write the inorder and postorder traversals of the tree given below 	2
Q2	A	Write an implementation of a queue of characters using an array and ONLY one index in the array. (Explanation: You have to write the type-definition and the functions <i>init()</i>, <i>enqueue()</i>, <i>dequeue()</i>, <i>empty()</i>; for the queue)	3
	B	Using the given type definition of a binary tree, write a recursive function which computes the average of all the numbers stored in the tree. (Do not assume to be a search tree). <pre> typedef struct node { int val; struct node *left, *right; }node; typedef struct node *tree; </pre>	3
	C	Draw the diagram of the hash table, using following hash function, and after the given insertions are made into it. Assume linear probing for collision resolution. <pre> int ht[7]; int hash(int n) { return (n - 1) % 7; } insertions: 20, 5, 19, 4, 3 </pre>	3
	D	Write a complete program which reads following structures written in a binary-file, into an array. Then the program writes the structures into a text-file. The names of the two files are given as command line arguments. <pre> typedef struct data { char name[16]; int age; int marks; }data; </pre>	3
	E	Write a function which creates transpose of a sparse matrix. The sparse matrix is stored using following type definition.	5

		<pre> void transpose(spm *s); typedef struct elem { int r, c, v; }elem; typedef struct spm { elem arr[128]; /* elements are stored sorted in (r,c) pairs */ int nrow, ncol, nelem; }spm; </pre>	
	F	What is the output of the following program? Explain in 2-3 lines. <pre> #include <stdio.h> #define A 1 #define B 2 #define C 4 #define D 8 int f(int a, int b) { if((a & A) && (a & B)) return A; if((a & B) && (a & C)) return B; if((a & D) && (a & C)) return C; return b; } int main() { printf("%d\n", f(A B, D)); printf("%d\n", f(B C D, A)); printf("%d\n", f(B D, C)); } </pre>	3
Q3	A	The C type 'int' has a limitation of size. Write an implementation of a new data type which gives us "integers of unlimited size", using a linked-list type of implementation. The type will be called "Integer". Write the following functions for your type: <pre> Integer *init(void); /* returns a new & empty variable of type Integer */ void add_digit(Integer *i, int x); /* Adds a new digit at the end of an existing integer */ Integer *add(Integer a, Integer b); /* Adds two integers a and b, and returns the result as another "Integer" */ </pre>	4
	B	Write a function which will print a binary tree so that it "looks like" a binary tree. Use the following typedefs: <pre> typedef struct node { char ch; struct node *left, *right; }node; typedef struct node *tree; </pre>	4

	C Write a function which given 'n' as base of a number system, and 'm' as no. of positions, prints all numbers in that number system having 'm' positions. Find the time complexity of your function. <i>Suppose the function is</i> <pre>void printall(int m, int n);</pre> <i>Then when called as printall(2, 2), it will print 00,01,10,11 (Numbers having 2 bits, in binary number system); when called as printall(2,3) it will print 00,01,02,10,11,12,20,21,22 (Numbers having 2 positions, in ternary number system).</i>	4
	D The C library has a function 'qsort' for sorting of data of any type. Write a function which imitates this function, using bubble sort. <i>Desired prototype:</i> <pre>void bsort(void *base, size_t nmemb, size_t size, int (*compare)(const void *, const void *));</pre> <i>The function sorts an array with 'nmemb' elements of size 'size'. The 'base' argument points to the start of the array. Here 'compare' is a function pointer, for a function which compares two data elements of the data to be sorted.</i>	4
	E Design data structures for an application which manages a library. Write the type definitions and draw relevant diagrams of the data structure. No need to write any explanation, your type definitions and diagrams should be self-sufficient.	4